Auditing and Improving FinRobot's Equity-Research Pipeline

STAT-GR5398 Assignment 1 - FinRobot Equity Research AI Agent Track

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Executive Summary

I evaluated FinRobot on five Technology-sector companies (NVDA, AMD, INTC, AAPL and MSFT), audited source data and generated claims, reproduced valuation logic, tested stability, and refactored the valuation layer. Filing-linked accounting values reconciled well for Apple and Microsoft, but narrative grounding was materially weaker: 142 of 261 audited claims were directly supported by the agents' shared evidence base (54.4%). The strongest section was Major Takeaways (100% grounded) and the weakest was Company Overview (22.8%). The valuation refactor replaced synthetic net debt, EBITDA×60% FCF, arbitrary fallbacks, and a broken peer-data path with actual cash flow, actual net debt, historical multiples, explicit DCF assumptions, common-year peers and deterministic synthesis. Stability testing also exposed a non-LLM failure: FMP rate limiting turned live rating/target fields into N/A/0 in two of three repeated NVDA runs.

1. Setup and Coverage Universe

Model: GPT-4.1. Data provider: Financial Modeling Prep (FMP); Google News RSS was used as a fallback when the FMP news endpoint returned 402. Adanos retail sentiment was not configured.

Coverage universe: NVIDIA, AMD, Intel, Apple and Microsoft, all classified by the report pipeline as Technology. Each company was run with at least two peers. The mix intentionally includes hypergrowth, mature-platform and turnaround profiles.

Exact API cost and end-to-end baseline runtime were not logged by the saved pipeline artifacts, so this report does not invent dollar or runtime values. The saved experiment contains 40 baseline GPT-4.1 narrative calls (8 sections × 5 companies) plus 16 calls from two additional NVDA stability reruns, or 56 narrative calls total. FMP cost depends on the user's subscription plan.

2. Data Audit Against SEC Filings

I reconciled 22 items across Apple and Microsoft. All selected items matched at the displayed precision. The main semantic issue is naming: FinRobot's 'Contribution Profit' is mathematically Revenue minus Cost of Operations and equals filing gross profit/gross margin.

Company	Metric	Year	Pipeline	Filing	Verdict
AAPL	Revenue	2025	416161	416161	Match
AAPL	Revenue	2024	391035	391035	Match
AAPL	Cost of Operations	2025	220960	220960	Match
AAPL	Cost of Operations	2024	210352	210352	Match
AAPL	SG&A	2025	27601	27601	Match
AAPL	SG&A	2024	26097	26097	Match
AAPL	Contribution Profit	2025	195201	195201	Match

AAPL	Contribution Profit	2024	180683	180683	Match
AAPL	Contribution Margin	2025	46.9	46.9	Match
AAPL	Revenue Growth	2025	6.4	6.4	Match
AAPL	EPS	2025	7.49	7.49	Match
MSFT	Revenue	2026	331839	331839	Match
MSFT	Revenue	2025	281724	281724	Match
MSFT	Cost of Operations	2026	106374	106374	Match
MSFT	Cost of Operations	2025	87831	87831	Match
MSFT	SG&A	2026	34666	34666	Match
MSFT	SG&A	2025	32877	32877	Match
MSFT	Contribution Profit	2026	225465	225465	Match
MSFT	Contribution Profit	2025	193893	193893	Match
MSFT	Contribution Margin	2026	67.9	67.9	Match
MSFT	Revenue Growth	2026	17.8	17.8	Match
MSFT	EPS	2026	18.0	18.0	Match

3. Forecast and Valuation Audit

The baseline forecasting logic is not a statistical forecasting model. It applies the same 5%/6%/4% revenue-growth sequence to each company. More seriously, the reported '+1.0% annual margin improvement' does not map consistently into forecast EBITDA margins.

Ticker	Latest actual EBITDA margin	First forecast margin	Change	Finding
NVDA	66.9%	70.5%	+3.6pp	Does not consistently equal +1.0pp
AMD	21.0%	39.0%	+18.0pp	Does not consistently equal +1.0pp
INTC	27.2%	27.6%	+0.4pp	Does not consistently equal +1.0pp
AAPL	34.7%	41.8%	+7.1pp	Does not consistently equal +1.0pp
MSFT	62.5%	59.0%	-3.5pp	Does not consistently equal +1.0pp

This matters because forecast EBITDA feeds EV/EBITDA and peer valuation. AMD is the clearest example: its EBITDA margin jumps from 21.0% actual to 39.0% forecast despite the printed +1.0pp assumption.

4. Grounding Audit

For NVDA, AAPL and MSFT I extracted factual/quantitative claims from all eight agent sections and classified them against the common evidence base (financial metrics, peer tables and saved recent news). Supported

means directly derivable; Unsupported means plausible but not established; Contradicted means the supplied evidence says otherwise.

Section	Supported	Unsupported	Contradicted	Total	Grounding rate
tagline	8	2	0	10	80.0%
company_overview	13	44	0	57	22.8%
investment_overview	30	15	0	45	66.7%
valuation_overview	26	10	2	38	68.4%
risks	9	7	0	16	56.2%
competitor_analysis	21	15	0	36	58.3%
major_takeaways	19	0	0	19	100.0%
news_summary	16	24	0	40	40.0%

Overall grounding rate: 142/261 = 54.4%. Major Takeaways performs best because it mostly restates supplied financial numbers. Company Overview performs worst because the shared prompt does not contain a full business profile, yet the agent introduces products, segments, partnerships and market-position claims. News Summary also contains unsupported synthesis when it moves beyond the saved headlines into causal explanations.

Two direct contradictions were identified in the valuation narrative: NVIDIA was described as remaining above AMD on EV/EBITDA even though NVDA 2025A (33.78x) was below AMD (47.85x); Microsoft's 2026A EV/EBITDA (13.87x) was described as closely aligned with Cisco's 23.66x.

5. Stability Test

Three nominally identical NVDA runs were preserved. The top-three risk headings were identical in all runs, so the narrative risk ranking was stable in this sample. However, Runs 2 and 3 hit FMP HTTP 429 limits on the live market-data endpoints; rating became N/A and target/valuation fields fell to zero. Raw target-price population variance including those zero sentinels is 25,563.6. A valid-value variance cannot be estimated because only Run 1 had a valid live target. This is an important reproducibility failure: external API availability is not controlled, and failure values are not cleanly separated from economic zeros.

6. Improvement: A More Auditable Valuation Engine

I chose Part 3 option C (a real valuation). The upstream engine used several shortcuts: a 12x EV/EBITDA fallback when the historical-multiple column did not match, synthetic net debt equal to 10% of enterprise value, FCF approximated as 60% of EBITDA, 10% growth for years 1-5 and 5% for years 6-10, and a peer-data interface that returned zero when given the report pipeline's DataFrame. The modified engine uses actual cash flow, actual debt minus cash, first-forecast-year EBITDA, historical median EV/EBITDA, a common peer fiscal year, explicit 10% WACC and 2.5% terminal growth, and equal weights across valid methods.

Ticker	Upstream reconstructed target	Modified target	Change	Modified methods
NVDA	$82.49	$184.52	$+102.03	EV + DCF + Peer
AMD	$111.97	$204.46	$+92.49	EV + DCF + Peer
INTC	$40.24	$77.57	$+37.33	EV + Peer
AAPL	$159.31	$198.09	$+38.78	EV + DCF + Peer
MSFT	$347.73	$471.33	$+123.60	EV + DCF + Peer

7. Independent NVDA Valuation Reproduction

The accompanying workbook `improvement/valuation_reproduction.xlsx` reproduces the modified NVIDIA DCF and EV/EBITDA logic from saved inputs. The DCF uses 2024-2026 actual FCF margins of 44.35%, 46.63% and 44.77%, for a normalized margin of 45.25%. Near-term FCF is forecast revenue multiplied by this normalized margin; years 4-5 grow 3%, years 6-10 grow 2.5%, WACC is 10%, terminal growth is 2.5%, and actual net debt is $0.807B. The engine's DCF target is $59.62 per share.

The wide disagreement across methods is economically meaningful: NVIDIA's historical-multiple method is $287.61, peer method $206.32, and DCF $59.62. The multiple methods capitalize near-term EBITDA at market multiples, while the DCF is constrained by normalized FCF conversion and a 10% discount rate. Therefore the synthesized $184.52 should be read as a model-combination output, not a statistical confidence estimate.

8. Sensitivity

The saved sensitivity analyzer moves revenue growth by -5.0 to +5.0 percentage points and EBITDA margin by -2.0 to +2.0 points. For NVIDIA, the 2029E combined EBITDA scenarios range from about $169.8B to $198.1B. To connect sensitivity to fair value, `audits/nvda_valuation_sensitivity.csv` applies the modified EV/EBITDA method to the first forecast year across the same growth/margin deltas. This makes the valuation consequence of forecast assumptions directly inspectable.

9. Judgment and Remaining Limitations

I would not hand the unreviewed report directly to a portfolio manager. Filing-derived financial tables are close to decision-grade after terminology cleanup and source labeling. The refactored valuation layer is auditable but still depends on simplistic forecast assumptions and, for AAPL/MSFT, imperfect peer comparability. The narrative layer is a draft-writing aid rather than a verified research product at a 54.4% grounding rate.

Remaining work should include: replacing hard-coded forecasts with a backtested forecasting model; caching and versioning all market data; treating API errors as missing values rather than zeros; adding a verifier agent; renaming Contribution Profit to Gross Profit; normalizing displayed method weights; and separating analyst consensus targets from model-derived targets.

References

Apple Inc. 2025 Form 10-K: https://www.sec.gov/Archives/edgar/data/320193/000032019325000079/aapl-20250927.htm

Microsoft Corp. 2026 Form 10-K: https://www.sec.gov/Archives/edgar/data/789019/000119312526323660/msft-20260630.htm

FinRobot upstream valuation engine: https://github.com/AI4Finance-Foundation/FinRobot/blob/master/finrobot_equity/core/src/modules/valuation_engine.py

Course assignment: https://github.com/AI4Finance-Foundation/STAT-GR5398-Columbia-University/tree/master/Fall-2026/Assignment1/FinRobot